

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

Software Engineering - Machine Learning

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

**BACHELOR OF ENGINEERING IN Department of
Artificial Intelligence and Machine Learning**

Submitted by:

Maaz Karim – 1DS20AI030

Conducted at



letsdive.io

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

2023-24

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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CERTIFICATE

This is to certify that the Internship titled **Machine Learning Intern** carried out by **Maaz Karim** bearing USN **1DS20AI030**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (20AI8ICINT)

Signature of Reporting Head

Kruthika K R

Tech Lead

Sandlogic

Signature of Internship Coordinator

Prof. Kavya D N

Dept. of AI & ML

DSCE

Signature of HoD

Dr. Vindhya P Malagi

Dept. of AI & ML

DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Maaz Karim**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College Of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, in **Dive**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023-2024.

Date:

USN: 1DS20AI030

Place: Bangalore

NAME: Maaz Karim

OFFER LETTER



Dive

/home/eyepatch/Desktop

Letsdive Technologies India Private
CIN: U72900DL2020PTC373643
Address: 602, Block A, Naurang House,
21K.G.Marg, Connaught Place, Delhi, 110001
Email: nitesh@letsdive.io

Offer Letter

Offer Letter for Joining **Letsdive Technologies India Private Limited** (the "Company")

Date: 31st Jul 2023

Dear Maaz Karim,

We are pleased to extend you a full-time offer to join Letsdive Technologies India Private Limited as **Software Engineer Intern – ML**, for a period of **3 months**. The terms of your relationship with the company will be as per Attachment A, annexed herewith.

Your salary will be **40,000/-** per month.

Your joining date would be **1st August 2023**. You are expected to join on or before the said date failing which this letter stands cancelled.

We look forward to your acceptance of the offer and a mutually beneficial work relationship.

Regards,
Nitesh Agrawal
Chief Executive Officer

COMPLETION CERTIFICATE



Letsdive Technologies India Private Limited

CIN: U72900DL2020PTC373643

Address: 602, Block A, Naurang House,
21 K.G. Marg, Connaught Place, Delhi, 110001

Email: hello@letsdive.io

5th Jan 2024

To whomsoever it may concern

This is to inform you that **Maaz Karim**, a student of Dayananda Sagar Institutions has been offered **Machine Learning Engineer Internship** with Dive during the period Aug 2023 - Jan 2024.

The Machine Learning Engineer Intern Program is a unique 6-month internship that offers personal, professional development in the field of AI and ML providing students with an opportunity to work on complex real world problems, develop scalable, distributed software systems, and also collaborate on multitudes of smaller projects that have universal appeal - which requires research, awareness, interactivity, and the ability to ask the right questions.

During the course of this Internship as a MLE Intern, Maaz had the opportunity to work on the critical systems that powers our AI pipelines and it's real time components. Maaz has a breadth of experience to become a great engineer in the future with expertise in AI/ML.

The Machine Learning Engineer Intern participates in the SDLC process in a full-fledged way while promoting best practices that we believe will pay dividends in the long term, such as fearlessly owning products end-to-end as we come across them rather than leaving them to future engineers.

In case of questions, please do not hesitate to reach out to us.

Regards

Om Prakash Shanmugam

Founder & CTO, Dive

om@letsdive.io

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal Dr. B G Prasad, for providing us adequate facilities to undertake this Internship.

We would like to thank Dr. Vindhya P Malagi, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank Kruthika K R, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Coordinator, Prof. Kavya D N, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of the AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

Maaz Karim
1DS20AI030

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CHAPTER 1

INTRODUCTION

My internship at Dive has been a pivotal experience, shaping my understanding of AI technologies and their practical applications. Dive's focus on AI-powered meeting automation solutions presented me with a challenging yet rewarding environment to expand my skills and knowledge. From building template-based summarization systems to optimizing real-time LLM inference pipelines, each project has been a stepping stone in my professional development.

The company's culture, characterized by a blend of talent, innovation, and respect, provided a supportive backdrop for learning. Dive's emphasis on continuous learning and collaboration encouraged me to push my limits and explore new possibilities. This report encapsulates my journey, highlighting key achievements, challenges faced, and recommendations for future interns based on my experiences.

Throughout my internship, I have had the privilege of working with a diverse team of talented individuals. This collaborative environment not only enriched my learning experience but also exposed me to different perspectives and approaches to problem-solving. I have learned the importance of teamwork and communication in achieving common goals, and how these skills are essential in a dynamic work environment.

One of the most rewarding aspects of my internship has been the opportunity to work on cutting-edge technologies like AI and machine learning. Dive's commitment to innovation has allowed me to delve deep into these technologies, expanding my knowledge and skills in areas such as NLP-based summarization and LLM optimization. These hands-on experiences have not only enhanced my technical capabilities but also broadened my understanding of AI's potential impact on various industries.

Moreover, my internship at Dive has taught me the importance of client-centricity and quality assurance. Understanding the needs of our clients and ensuring that our products meet their expectations has been a key focus area. I have learned how to prioritize client-related features and how to effectively test the product to ensure its quality and reliability. These experiences have provided me with valuable insights into the importance of customer satisfaction and product excellence.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Dive is an innovative AI-powered meeting assistant that revolutionizes the way you conduct and manage meetings. With Dive, you can automate note-taking, create seamless meeting summaries, and track action items effortlessly.

Key Features:

- **Automated Notes:** Dive's AI assistant takes care of note-taking, allowing you to focus on discussions.
- **Action Items:** Automatically generated summaries and action items make meetings more actionable.
- **Effortless Search:** Easily find key insights and decisions from past meetings.
- **Integrated Workflow:** Seamlessly integrates with existing tools like Google Docs, Notion, and ClickUp.
- **Security & Privacy:** Dive ensures enterprise-grade security and GDPR compliance for peace of mind.

For 1-1 meetings, Dive helps spark engaging conversations, nurture collaboration, and inspire personal growth. It transforms one-on-ones into game-changers by fostering meaningful connections and driving productivity.

For team meetings, Dive empowers teams to boost revenue growth with AI-driven meetings that enhance collaboration, simplify cross-department coordination, and speed up projects. It offers pre-made templates designed to make meetings more efficient, ensuring meaningful conversations and the best use of everyone's time.

CHAPTER 3

SCOPE OF WORK

During my tenure at Dive, I've been deeply involved in several key projects aimed at revolutionizing meeting management and enhancing the utility of AI-driven tools for our users. One of my primary responsibilities involved building a system for template-based summarization from meeting transcripts. This system has been instrumental in streamlining the summarization process, allowing for more efficient and accurate generation of meeting summaries. In addition to this, I've also played a key role in developing a pipeline for evaluating NLP-based summarization techniques. This pipeline has been crucial in helping us continually improve the performance of our language models, ultimately leading to more accurate and relevant summaries for our users.

Another significant project I've worked on is enhancing the output relevance in the AI summary of all Ask-AI features. This involved fine-tuning our language models and implementing new algorithms to ensure that the summaries provided by our AI assistant are not only accurate but also highly relevant to the user's needs. Additionally, I've developed a method for capturing and summarizing meeting highlights and a technique for automatically chapterizing meetings. Our users have highly praised these features for their ability to improve the organization and accessibility of meeting content.

In terms of technical contributions, I have trained and fine-tuned large language models (LLMs) such as LLaMa, leveraging llama-index to further improve our Retrieval Augmentation Generation (RAG). Additionally, I've utilized Langfuse to track and evaluate the performance of our LLMs, ensuring that they meet our high standards for accuracy and reliability. These tools have been instrumental in helping us maintain a high level of performance and deliver exceptional value to our users.

One of the most challenging yet rewarding projects I've undertaken at Dive is the optimization of the pipeline for real-time LLM inference. This involved identifying bottlenecks in the pipeline and implementing optimizations to improve its speed and efficiency. The optimized pipeline has significantly reduced the time it takes to process meeting data, allowing for faster and more responsive AI-driven features.

Furthermore, I've designed and built a proof of concept (POC) to allow contextual search across meetings. This feature has been a game-changer for our users, allowing them to easily search for specific topics or discussions within meeting transcripts. This has greatly improved the overall usability of our platform and has been well-received by our user base.

My work at Dive has been focused on pushing the boundaries of what is possible with AI-driven meeting management tools. I've been fortunate to work on projects that have not only challenged me technically but have also had a significant impact on the user experience. I look forward to continuing to innovate and drive positive change in the field of AI-driven meeting management.

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

As a Machine Learning intern at Dive, I was given specific learning objectives to enhance my skills and knowledge in key areas relevant to our work. One of the primary objectives was to deepen my understanding of deep learning and how Large Language Models (LLMs) work. This involved conducting extensive research on LLMs, including how they are evaluated and ways they can be fine-tuned for optimal performance.

Another important objective was to explore the various ways in which LLMs can be used, particularly focusing on Retrieval Augmentation Generation (RAG) techniques. This involved understanding the critical role that LLMs play in enhancing the quality of responses generated by AI systems and identifying key strategies for leveraging LLMs in different contexts.

Additionally, I was tasked with benchmarking both open-source and proprietary LLMs to understand their strengths, weaknesses, and performance metrics. This objective aimed to provide me with a comprehensive overview of the current landscape of LLMs and their applications in real-world scenarios.

Another learning objective was to deepen my knowledge of Django, a high-level Python web framework, with a focus on writing tests and integrating with GitHub Actions for continuous integration and deployment (CI/CD). This objective aimed to improve my ability to develop and maintain web applications using Django and ensure that our projects adhere to best practices for software development.

Furthermore, I was required to learn how to scale LLMs effectively to handle large volumes of data and requests, as well as understand how LLM-based agents work in AI systems. This objective aimed to enhance my understanding of the practical challenges and considerations involved in deploying LLM-based solutions in production environments.

Overall, these learning objectives provided me with valuable insights and skills that have been instrumental in my work at Dive, enabling me to contribute effectively to our projects and drive innovation in the field of AI-driven meeting management.

CHAPTER 5

CHALLENGES AND SOLUTION

CHALLENGES

Navigating the complexities of Large Language Models (LLMs) required extensive research and collaboration to optimize their performance. Integrating Django with GitHub Actions for CI/CD demanded thorough learning and implementation of best practices. Scaling LLMs for increased workload involved exploring distributed computing and parallel processing techniques. Understanding practical LLM applications in meeting management and AI-driven features required innovative use case identification and effective implementation. Balancing security and performance in AI-driven features involved implementing robust security measures and conducting regular audits. Adapting to changing requirements and technologies necessitated maintaining flexibility and regularly updating skills through continuous learning

SOLUTION

Thorough Understanding: Study latest papers for LLM's and documentation of Django to grasp core concepts.

Hands-On Practice: Engage in coding exercises and small projects to reinforce learning and gain practical experience.

Seek Mentorship: Consult experienced developers or mentors for guidance and assistance in navigating challenges.

Debugging Skills: Develop strong debugging techniques to identify and resolve issues effectively during project development.

Effective Time Management: Prioritize tasks and allocate time efficiently to ensure progress and meet deadlines.

Continuous Learning: Stay updated with the latest advancements in LLM's through ongoing learning and exploration.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

In my role at Dive, I have had the opportunity to contribute to several key projects and initiatives, leading to significant achievements that have positively impacted the company. My contributions have spanned various areas, including AI-driven meeting management, software development, and research. Here are some of the notable achievements and contributions during my time at Dive:

1. **Template-Based Summarization:** I played a crucial role in developing a system for template-based summarization from meeting transcripts. This system has improved the efficiency and accuracy of generating meeting summaries, enhancing the overall user experience.
2. **Enhanced Output Relevance:** I focused on improving the output relevance in the AI summary of all Ask-AI features. Through my efforts, we were able to deliver more relevant and useful summaries to our users, increasing the overall value of our AI assistant.
3. **Meeting Highlights and Chapterization:** I developed methods for capturing and summarizing meeting highlights, as well as creating automatic chapterization of meetings. These features have improved the organization and accessibility of meeting content, making it easier for users to find key information.
5. **Real-Time LLM Inference Optimization:** I optimized the pipeline for real-time LLM inference, leading to faster and more efficient processing of meeting data. This has improved the performance of our AI-driven features and ensured a smoother user experience.
6. **Contextual Search Across Meetings:** I designed and built a proof of concept (POC) to allow contextual search across meetings. This feature has made it easier for users to search for specific topics or discussions within meeting transcripts, improving overall usability.
7. **Deep Learning and LLM Research:** I conducted extensive research on deep learning and how LLMs work. This has helped me better understand the complexities of LLMs and how to optimize their performance for various applications.
8. **Django and GitHub Actions Mastery:** I deepened my knowledge of Django, including writing tests and integrating with GitHub Actions for CI/CD. This has improved our development workflow and ensured the reliability of our web applications.
9. **Scaling LLMs and LLM-Based Agents:** I learned how to scale LLMs effectively to handle large volumes of data and requests. Additionally, I gained an understanding of how LLM-based agents work, which has been valuable in developing and deploying AI-driven features.

CHAPTER 7

REFLECTION AND ANALYSIS

Reflecting on my experiences at Dive, I have not only expanded my technical skills but also developed a deeper understanding of the importance of features and client-related aspects in product development. One significant aspect of my growth has been learning how to give importance to features and think of new features to improve the platform independently. Through this process, I have honed my ability to identify user needs and translate them into innovative features that enhance the user experience.

Additionally, I have learned the importance of giving emphasis to client-related features. Understanding the needs and preferences of our clients has been instrumental in shaping the direction of our projects and ensuring that our products meet the needs of our users. By focusing on client-related features, I have been able to contribute to the development of solutions that are not only technically sound but also align with the goals and objectives of our clients.

Another valuable lesson I have learned at Dive is the importance of testing the product and features rigorously. Testing plays a crucial role in ensuring the quality and reliability of our products, and I have learned various testing methodologies and techniques to ensure that our products meet the highest standards. By emphasizing testing, I have been able to identify and address issues early in the development process, ultimately improving the overall quality of our products.

Overall, my experiences at Dive have been invaluable in helping me develop as a professional and as an individual. I am grateful for the opportunities I have had to learn and grow, and I look forward to applying the skills and knowledge I have gained in future endeavors.

CHAPTER 8

RECOMMENDATIONS

Working as an intern at Dive is a unique and enriching experience that offers a blend of learning, growth, and meaningful contributions. The company's culture, characterized by a chill environment and a high level of respect for everyone, creates a welcoming and collaborative atmosphere. This culture is conducive to learning and innovation, making it an ideal environment for interns to thrive. The nature of work at Dive is challenging yet rewarding, with interns given the opportunity to work on cutting-edge technologies and collaborate with talented individuals. The company expects interns to contribute meaningfully to projects, embracing the values of respect, collaboration, and innovation.

One of the key recommendations for interns at Dive is to embrace the company culture. By aligning with the values of respect, collaboration, and innovation, interns can contribute positively to the team dynamic and make a meaningful impact on the company's projects. Another recommendation is to focus on continuous learning. Dive offers interns the opportunity to work with cutting-edge technologies and talented individuals, providing a unique learning experience. Interns should take advantage of this opportunity by actively seeking out new learning opportunities and challenging themselves to grow professionally.

Contribution is another key aspect of the internship experience at Dive. Interns are expected to take ownership of their work and strive to deliver high-quality results that align with the company's goals. Seeking feedback is also important, as it allows interns to improve their skills and performance. Interns should not be afraid to ask for feedback from colleagues and supervisors, and should use this feedback to grow as professionals.

Being proactive is another important recommendation for interns at Dive. By taking initiative and identifying areas where they can contribute and add value, interns can stand out and make a positive impact on the company's projects. Collaboration is also key at Dive, and interns should work closely with colleagues to achieve common goals. Maintaining a positive attitude is essential, as the work environment at Dive is known for being chill and respectful.

Challenges are an integral part of the internship experience at Dive, but they are also opportunities for growth. Interns should not shy away from challenges, but should instead see them as opportunities to develop their skills and knowledge. Striving for excellence is another important recommendation for interns at Dive. The company expects interns to deliver excellent results and contribute to the overall success of the company, and interns should aim for excellence in everything they do.

CHAPTER 9

CONCLUSION

Starting as an intern at Dive, I was immediately struck by the company's unique and vibrant culture. The environment was not only chill and respectful but also incredibly welcoming, making it easy to integrate into the team and feel like a valued member from day one. This culture of inclusivity and collaboration was instrumental in shaping my internship experience and setting the stage for a rewarding journey of learning and growth.

One of the most valuable aspects of my internship at Dive was the opportunity to work with cutting-edge technologies and talented individuals. From building template-based summarization systems to optimizing real-time LLM inference pipelines, I was constantly challenged to push the boundaries of what I thought was possible. This hands-on experience not only deepened my technical skills but also broadened my understanding of how AI and machine learning can be applied to real-world problems.

Another highlight of my internship was the emphasis on continuous learning and personal development. Dive provided a supportive environment where I was encouraged to explore new ideas, take on new challenges, and expand my skill set. Whether it was learning Django in depth or mastering the art of testing product features, Dive provided the resources and guidance I needed to grow both personally and professionally.

One of the key takeaways from my internship at Dive was the importance of giving importance to features and thinking of new ways to improve the platform independently. This mindset of innovation and continuous improvement was ingrained in the company culture, and it was inspiring to see how everyone, regardless of their role or experience level, was encouraged to think creatively and push the boundaries of what was possible.

Furthermore, I learned how to give emphasis on client-related features and how to test the product and features effectively. Dive's commitment to delivering exceptional products and services to its clients was evident in everything we did, and I was fortunate to be part of a team that prioritized client satisfaction above all else. Whether it was implementing robust security measures or conducting thorough testing of our AI-driven features, I learned firsthand the importance of putting the client first in everything we do.

Overall, my internship at Dive was a transformative experience that exceeded all my expectations. I not only gained valuable skills and knowledge but also made lifelong connections and memories. The company's commitment to excellence, innovation, and collaboration has left a lasting impression on me, and I am grateful for the opportunity to have been part of such an incredible team. As I move forward in my career, I will carry the lessons I learned at Dive with me, confident that they will serve me well in whatever challenges lie ahead.

CHAPTER 10

REFERENCES

1. <https://www.letsdive.io/>
2. <https://app.letsdive.io/>
3. <https://education.github.com/>

